

First Field Notes: A Random Embedded Systems Test Post

Peace Umolu

2026-08-02

[Download PDF](#)

Why this test post exists

This post is a quick test to verify that Quarto rendering works and the custom style sheet is being applied correctly.

If this page renders with:

- blue title accents,
- rounded cards/callouts,
- cyan buttons,
- styled code blocks,

then the theme hookup is successful.

Tiny project story

I connected a Cortex-M33 board to a sensor breakout and logged temperature values at fixed intervals.

Three immediate lessons:

1. Clock setup and peripheral init order matter more than expected.
2. Small buffer design choices can reduce dropped samples.
3. Good logging makes debugging ten times easier.

Measurement model and formula

For this run, I use a moving average filter where $x[n]$ is the latest raw ADC reading and N is the filter window:

$$x_{avg}[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[n-k]$$

When I estimate voltage from ADC counts for a 16-bit converter:

$$V_{in} = \frac{ADC_{count}}{2^{16} - 1} \cdot V_{ref}$$

These two equations make it easy to reason about noise reduction and conversion scaling before adding closed-loop control.

Data flow graphic



Bench image



Figure 1: ADC bring-up test bench graphic

The capture above represents the same measurement chain used for this test post: sensor input, sampled waveform, and firmware-side verification.

Sample data table

Time (s)	Temperature (C)	Status
0	24.8	Boot
5	25.1	Stable
10	25.4	Stable
15	26.0	Warm-up

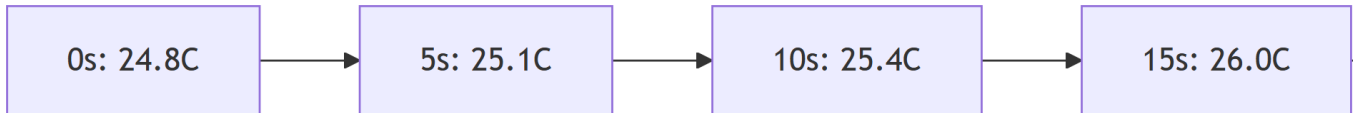
Example code chunk

```
# Simple moving average test for sensor readings
readings = [24.8, 25.1, 25.4, 26.0, 25.7, 25.5]
window = 3

ma = []
for i in range(len(readings) - window + 1):
    segment = readings[i:i + window]
```

```
ma.append(round(sum(segment) / window, 2))  
print("Moving averages:", ma)
```

Quick trend visualization



Practical next step

I will extend this test by:

- adding SPI transfer timing checks,
- introducing interrupt-driven sampling,
- and plotting results in the next journal entry.

i Quick check

If this callout appears in a rounded white panel, style.css is loading correctly for Quarto pages.